

13.8.8 Connection possibilities

Channels 1 to 4 can be connected as follows:

Channel	Function					
1	I	Event counter	A	A	SSI data	
2	I/O	Event counter	B	B	SSI clock	PWM
3	I	Event counter	A	R		
4	I/O	Event counter	B	Enable reference		PWM

The functions can also be mixed. For example:

Example 1	
Channel	Function
1	SSI data
2	SSI clock
3	Event counter
4	PWM

Example 2	
Channel	Function
1	SSI data
2	SSI clock
3	A
4	B

Example 3	
Channel	Function
1	Event counter
2	PWM
3	Event counter
4	PWM

Example 4	
Channel	Function
1	A
2	B
3	R
4	Enable reference

Example 5	
Channel	Function
1	A
2	B
3	Event counter
4	PWM

Example 6	
Channel	Function
1	Event counter
2	PWM
3	A
4	B

13.8.9 Input circuit diagram

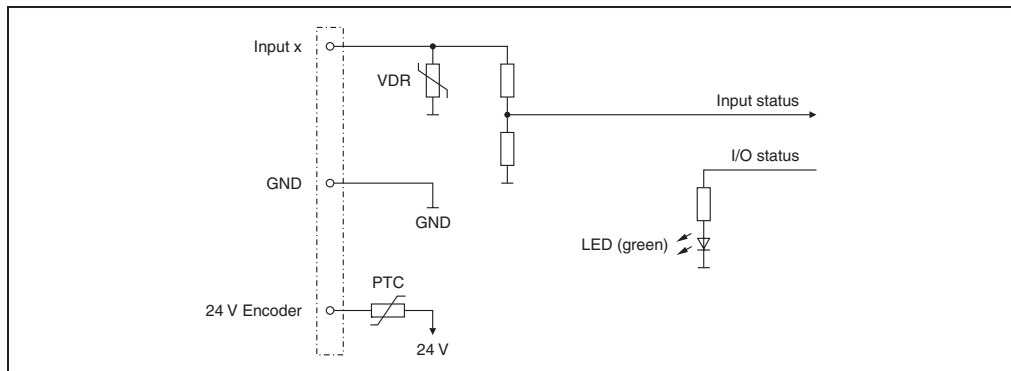


Figure 177: DC2395 - input circuit diagram

13.8.10 Output circuit diagram

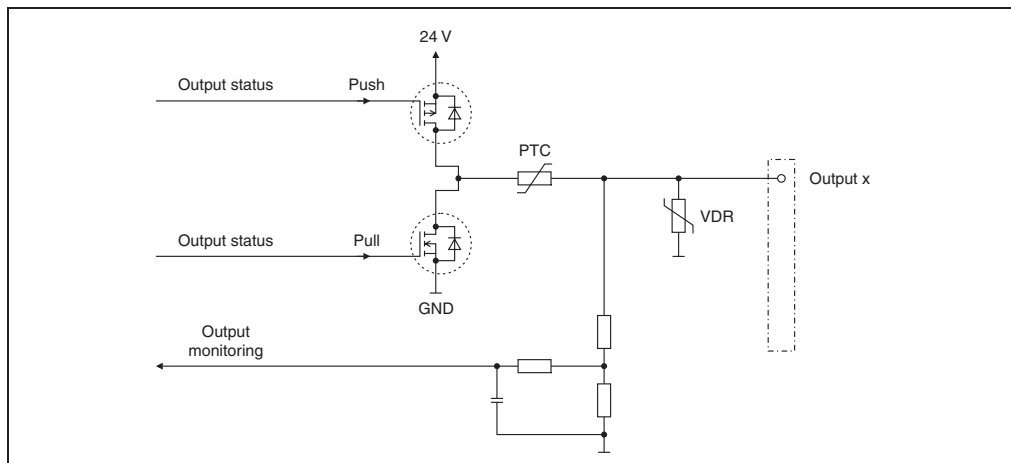


Figure 178: DC2395 - output circuit diagram

13.8.11 Switching inductive loads

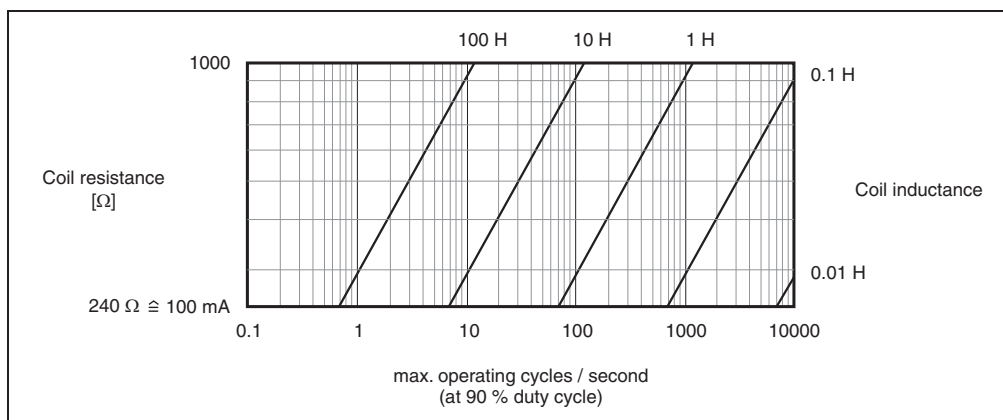


Figure 179: DC2395 - switching inductive loads

13.8.12 ReferenceModeEncoder01 channel

The reference mode is written cyclically by incremental encoder 1. Bit 0 and bit 1 make up the definition:

- \$C1 ... Single reference
- \$C3 ... Continual references

After a successful single reference (indicated by the status bits), the reference mode must first be turned back off (value = \$C0).

After the status has taken on the value \$00, the reference process can be started again with the value \$C1.

Bit	Description
0 - 1	Setting the reference mode 00 ... Off (default) 01 ... Single reference 10 ... Reserved 11 ... Continual references
2 - 5	0
6 - 7	1

13.8.13 StatusInput01 channel

The status of incremental encoder 1 is read cyclically.

Bit	Description
0 - 1	0
2	After the first reference pulse, this is always 1 0 ... Referencing is turned off 1 ... After the first reference pulse
3	Toggles after every completed reference 0 ... Referencing is turned off 0/1 ... Toggle after each completed reference
4	After the first reference pulse, this is always 1 0 ... Referencing is turned off 1 ... After the first reference pulse
5 - 7	Cyclic counter which is increased by 1 as soon as a reference pulse is detected 0 ... Referencing is turned off xxx ... Counter status

13.9 DC4395

13.9.1 General information

The DC4395 module is a multifunctional counter module. It can be connected to two SSI encoders, two ABR encoders, four AB encoders, or eight event counters. Four outputs are available for pulse width modulation. The functions can also be mixed (see section 13.9.8 "Connection possibilities" on page 411).

- 24 VDC encoder inputs
- SSI, ABR, AB or event counters for inputs
- Pulse width modulation for outputs
- 24 VDC and GND for encoder supply

13.9.2 Order data

Model number	Short description	Image
	Counter module	
X20DC4395	X20 digital counter module, 2 x SSI absolute encoders 24 V, 2 x ABR incremental encoder 24 V, 4 x AB incremental encoder 24 V, 8 x event counter, or 4 x PWM	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 223: DC4395 - order data

13.9.3 Technical data

Product ID	DC4395
Short description	
I/O module	2 SSI absolute encoder 24 V, 2 ABR incremental encoder 24 V, 4 AB incremental encoder 24 V, 8 x event counter, or 4 x pulse width modulation
SSI absolute encoder	
Number	2
Encoder inputs	24 V, asymmetrical
Counter size	32-bit
Maximum transfer rate	125 kBit/s
Encoder supply	Module-internal, max. 600 mA
Incremental encoder	
Number	4
Encoder inputs	24 V, asymmetrical
Counter size	16-bit
Input frequency (max.)	100 kHz
Evaluation	4-fold
Encoder supply	Module-internal, max. 600 mA
Event counter	
Number	8
Rated voltage	24 VDC
Counter size	16-bit
Evaluation	Each edge, cyclic counter
Signal form	Square wave pulse
Digital outputs	
Number	4
Rated voltage	24 VDC
Rated output current	0.1 A
Total current	0.4 A
Output circuit	Sink or source
Output protection	Thermal cutoff for over-current and short circuit, integrated protection for switching inductances.
Pulse width modulation ¹⁾	
Period duration	41.6 µs ... 500 ms
Pulse length	0 to 100 %
Resolution	0.1%
Actuator supply	Module-internal, max. 600 mA

Table 224: DC4395 - technical data

Product ID	DC4395
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error Outputs	Yes, with status LED and software status Yes, with status LED and software status (output status)
Electrical isolation Encoder - Bus Output - Bus Output - Encoder Encoder - Encoder Output - Output	Yes Yes No No No
Power consumption Bus I/O internal	Typ. 0.01 W Typ. 1.5 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 224: DC4395 - technical data (cont.)

1) Dead time when switching between push and pull: Max. 1.5 µs

13.9.4 Additional technical data

Product ID	DC4395
SSI absolute encoder	
Coding	Gray/Binary
CLK: Output current	Max. 100 mA
Overload behavior of the encoder supply	Short-circuit protection, overload protection

Table 225: DC4395 - additional technical data

Product ID	DC4395
Incremental encoder	
Overload behavior of the encoder supply	Short-circuit protection, overload protection
Event counter	
Input frequency	100 kHz
Count frequency	200 kHz
Input filter Hardware Software	$\leq 2 \mu\text{s}$ -
Input current at 24 VDC	Approx. 1.3 mA
Input resistance	18.4 k Ω
Switching threshold Low High	<5 VDC >15 VDC
Isolation voltage betw. channel and bus	500 V _{eff}
Digital outputs	
Type	Push / Pull / Push-Pull
Switching voltage	24 VDC (-15 % / +20 %)
Diagnostics status	Output monitoring
Leakage current when switched off	Max. 25 μA
Residual voltage	<0.9 V @ 0.1 A rated current
Short circuit peak current	<10 A
Switching on after overload or short circuit cutoff	approx. 10 ms (depends on the module temperature)
Switching delay 0 → 1 1 → 0	<2 μs <2 μs
Switching frequency Resistive load Inductive load	Max. 24 kHz See section 13.9.11 "Switching inductive loads" on page 413 (with 90 % duty cycle)
Braking voltage when switching off inductive loads	Switching voltage + 0.6 VDC
Isolation voltage betw. channel and bus	500 V _{eff}
General information	
B&R ID code	\$1CC5
Input voltage	24 VDC (-15 % / +20 %)
Type of signal lines	Shielded cables should be used for all signal lines

Table 225: DC4395 - additional technical data (cont.)

13.9.5 Status LEDs

Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			On	Error or reset state
	1 - 8	Green		Status of the corresponding digital signal

Table 226: DC4395 - status indicators

13.9.6 Pin assignments

Shielded cables should be used for all signal lines.

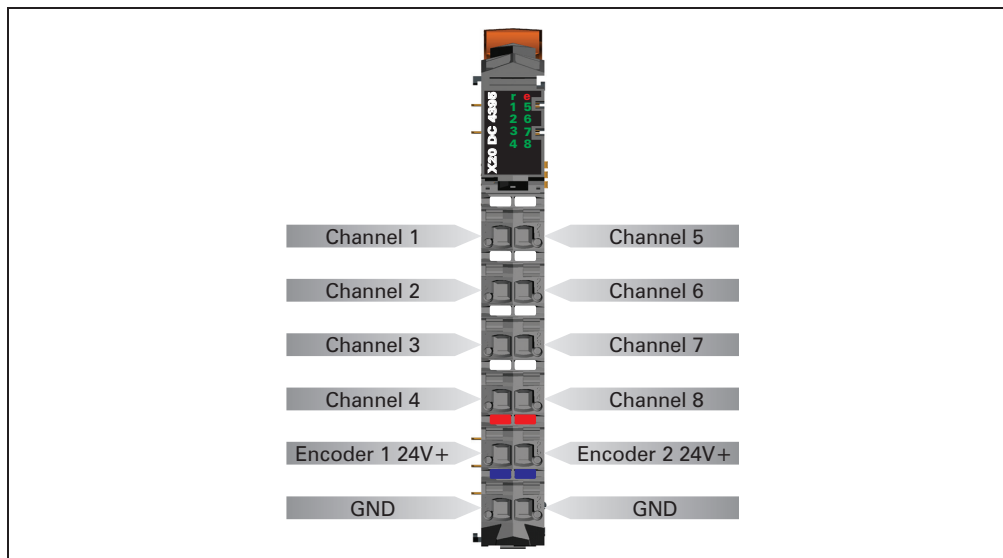


Figure 180: DC4395 - pin assignments

13.9.7 Connection example

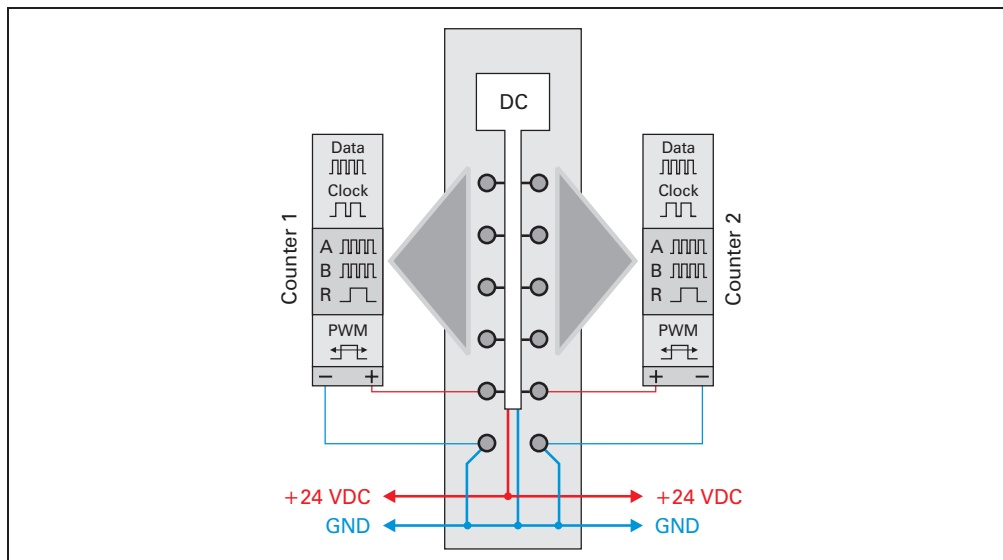


Figure 181: DC4395 - connection example

13.9.8 Connection possibilities

Channels 1 to 8 can be connected as follows:

Channel	Function					
1	I	Event counter	A	A	SSI data	
2	I/O	Event counter	B	B	SSI clock	PWM
3	I	Event counter	A	R		
4	I/O	Event counter	B	Enable reference		PWM
5	I	Event counter	A	A	SSI data	
6	I/O	Event counter	B	B	SSI clock	PWM
7	I	Event counter	A	R		
8	I/O	Event counter	B	Enable reference		PWM

The functions can also be mixed. For example:

Example 1	
Channel	Function
1	SSI data
2	SSI clock
3	Event counter
4	PWM
5	A
6	B
7	Event counter
8	PWM

Example 2	
Channel	Function
1	SSI data
2	SSI clock
3	A
4	B
5	Event counter
6	Event counter
7	Event counter
8	Event counter

Example 3	
Channel	Function
1	Event counter
2	PWM
3	Event counter
4	PWM
5	SSI data
6	SSI clock
7	A
8	B

Example 4	
Channel	Function
1	A
2	B
3	R
4	Enable reference
5	A
6	B
7	R
8	Enable reference

Example 5	
Channel	Function
1	A
2	B
3	Event counter
4	PWM
5	A
6	B
7	Event counter
8	Event counter

Example 6	
Channel	Function
1	Event counter
2	Event counter
3	Event counter
4	PWM
5	SSI data
6	SSI clock
7	A
8	B

13.9.9 Input circuit diagram

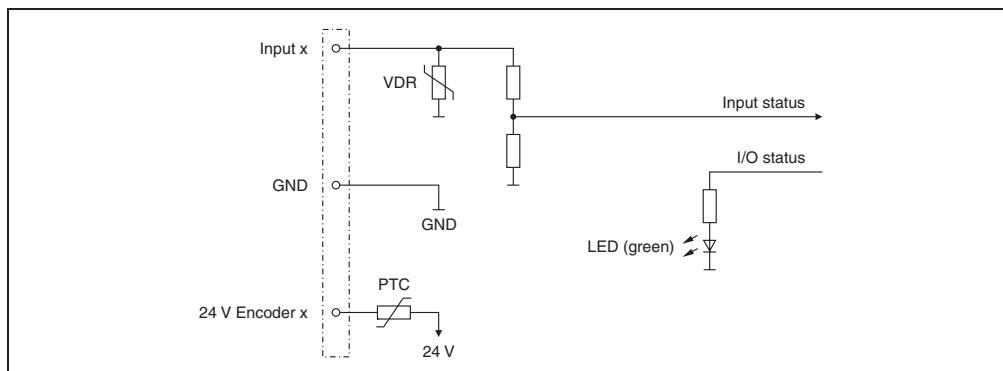


Figure 182: DC4395 - input circuit diagram

13.9.10 Output circuit diagram

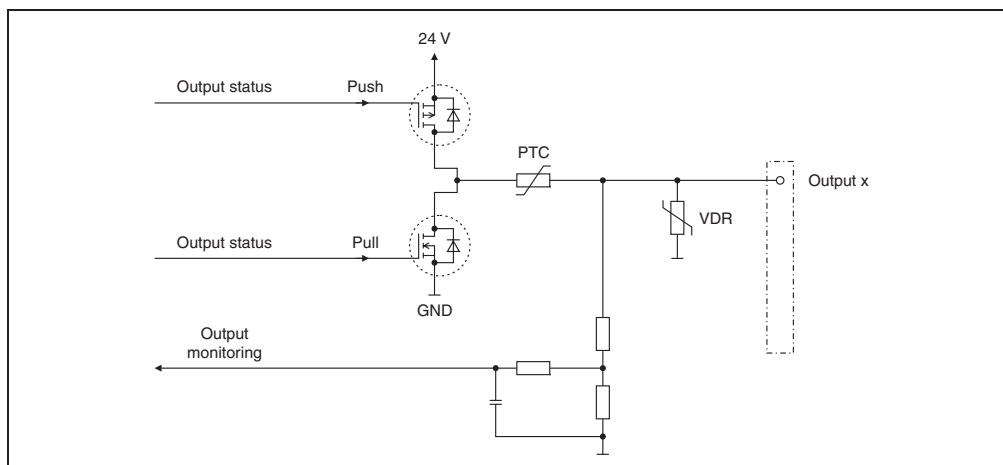


Figure 183: DC4395 - output circuit diagram

13.9.11 Switching inductive loads

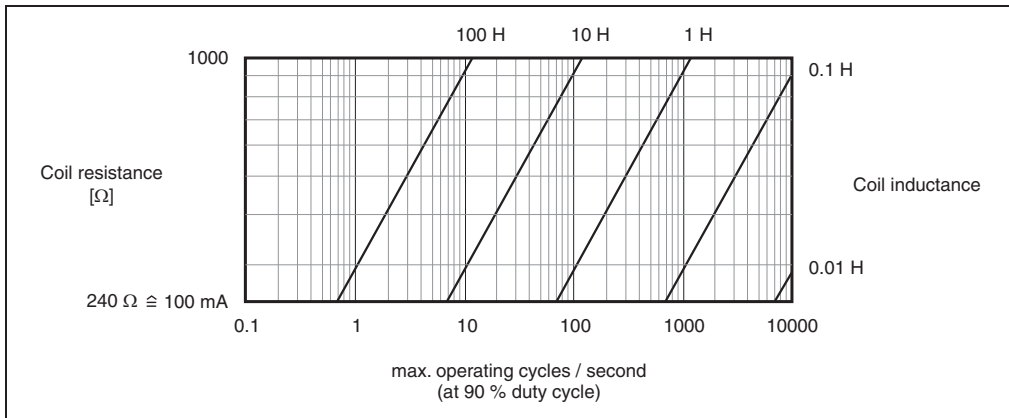


Figure 184: DC4395 - switching inductive loads

13.9.12 ReferenceModeEncoder01 channel

The reference mode is written cyclically by incremental encoder 1. Bit 0 and bit 1 make up the definition:

- \$C1 ... Single reference
- \$C3 ... Continual references

After a successful single reference (indicated by the status bits), the reference mode must first be turned back off (value = \$C0).

After the status has taken on the value \$00, the reference process can be started again with the value \$C1.

Bit	Description
0 - 1	Setting the reference mode 00 ... Off (default) 01 ... Single reference 10 ... Reserved 11 ... Continual references
2 - 5	0
6 - 7	1

13.9.13 ReferenceModeEncoder02 channel

The reference mode is written cyclically by incremental encoder 2. Bit 0 and bit 1 make up the definition:

- \$C1 ... Single reference
- \$C3 ... Continual references

After a successful single reference (indicated by the status bits), the reference mode must first be turned back off (value = \$C0).

After the status has taken on the value \$00, the reference process can be started again with the value \$C1.

Bit	Description
0 - 1	Setting the reference mode 00 ... Off (default) 01 ... Single reference 10 ... Reserved 11 ... Continual references
2 - 5	0
6 - 7	1

13.9.14 StatusInput01 channel

The status of incremental encoder 1 is read cyclically.

Bit	Description
0 - 1	0
2	After the first reference pulse, this is always 1 0 ... Referencing is turned off 1 ... After the first reference pulse
3	Toggles after every completed reference 0 ... Referencing is turned off 0/1 ... Toggle after each completed reference
4	After the first reference pulse, this is always 1 0 ... Referencing is turned off 1 ... After the first reference pulse
5 - 7	Cyclic counter which is increased by 1 as soon as a reference pulse is detected 0 ... Referencing is turned off xxx ... Counter status

13.9.15 StatusInput02 channel

The status of incremental encoder 2 is read cyclically.

Bit	Description
0 - 1	0
2	After the first reference pulse, this is always 1 0 ... Referencing is turned off 1 ... After the first reference pulse
3	Toggles after every completed reference 0 ... Referencing is turned off 0/1 ... Toggle after each completed reference
4	After the first reference pulse, this is always 1 0 ... Referencing is turned off 1 ... After the first reference pulse
5 - 7	Cyclic counter which is increased by 1 as soon as a reference pulse is detected 0 ... Referencing is turned off xxx ... Counter status

14. Dummy module

14.1 ZF0000

14.1.1 General information

The ZF0000 module is used as a place holder for later system expansion.

- Place holder for later system expansion
- Used as a terminal holder
- Module with no electrical function

14.1.2 Order data

Model number	Short description	Image
	Dummy module	
X20ZF0000	Dummy X20 module (non-functional)	
	Required accessories	
X20TB06	Standard X20 terminal block (6-pin)	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM01	X20 supply bus module, internal I/O supply is split to the left	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 227: ZF0000 - order data

14.1.3 Technical data

Product ID	ZF0000
Short description	
Accessories	Non-functional dummy module
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB06 or X20TB12 separately Order standard terminal block 1 x X20BM11 or supply bus module 1 x X20BM01 separately

Table 228: ZF0000 - technical data

14.1.4 Pin assignments

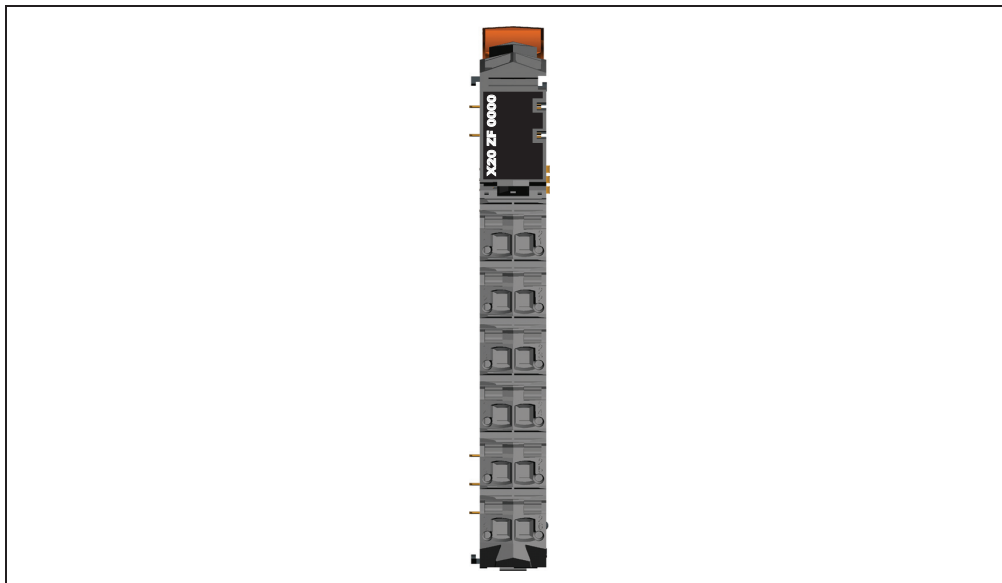


Figure 185: ZF0000 - pin assignments

14.1.5 Connection example

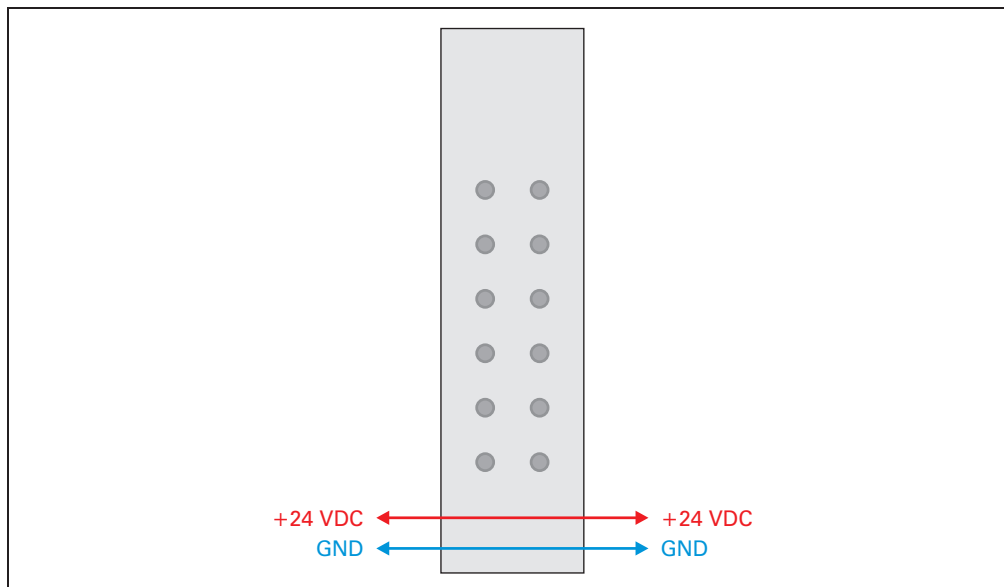


Figure 186: ZF0000 - connection example

Chapter 5 • Accessories

1. Additional equipment for X20 modules

Accessories available for the X20 modules include a plain text cover, an additional locking clip, and a cover holder.

Installation of these accessories is described in chapter 6 "Mechanical design", section 6 "Installation of accessories" on page 442.

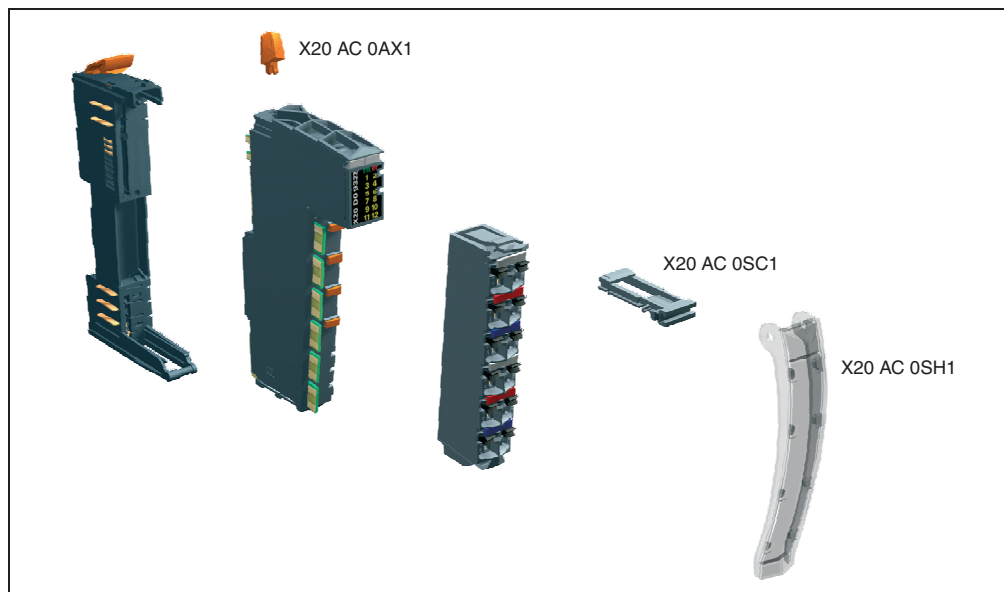


Figure 187: Additional equipment for X20 modules

1.1 Cover holder, terminal locking clip

Model number	Short description	Image
X20AC0SC1	X20 terminal locking clip and cover holder for plain text cover.	
X20AC0SC1.0100	X20 terminal locking clip and cover holder for plain text cover, 100 unit package	

1.2 Plain text cover

Model number	Short description	Image
X20AC0SH1	X20 plain text cover	
X20AC0SH1.0100	X20 plain text cover, 100 unit package	
X20AC0LB1.0100	X20 legend strips for X20 plain text cover, white paper, perforated, 96 strips on A4 sheets, 100 sheets per package	

1.3 Additional locking clip

Model number	Short description	Image
X20AC0AX1	X20 additional locking clip	
X20AC0AX1.0100	X20 additional locking clip, 100 unit package	

2. Locking plate

The locking plate protects the end modules from dirt and damage.

Model number	Short description	Image
X20AC0SL1	X20 locking plate (left)	
X20AC0SR1	X20 locking plate (right)	
X20AC0SL1.0010	X20 locking plate (left), 10 unit package	
X20AC0SR1.0010	X20 locking plate (right), 10 unit package	

3. Printer

This printer is used for printing the label tabs for additional terminal identification.

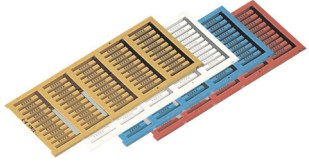
The ink cartridge is not included in the delivery.

Model number	Short description	Image
X20AC0PR1	X20 Ink Jet printer, special printer for X20 identification system. Order X20AC0CB1 ink cartridge separately.	
X20AC0CB1	X20 printer cartridge (black)	

4. Terminal labeling

Each terminal is clearly labeled directly on the terminal. Additional label tabs can be applied to further identify the terminals.

The labeling tool is required to apply these tabs (see also chapter 6 "Mechanical design", section 7 "Label tabs" on page 447).

Model number	Short description	Image
	X20 label tabs (blank)	
X20AC0M01	Blank X20 label tabs, white, set for 16 modules	
X20AC0M02	Blank X20 label tabs, red, set for 16 modules	
X20AC0M03	Blank X20 label tabs, blue, set for 16 modules	
X20AC0M04	Blank X20 label tabs, orange, set for 16 modules	
	X20 label tabs (printed)	
X20AC0M11	Printed X20 label tabs, white, set for 16 modules	
X20AC0M12	Printed X20 label tabs, red, set for 16 modules	
X20AC0M13	Printed X20 label tabs, blue, set for 16 modules	
X20AC0M14	Printed X20 label tabs, orange, set for 16 modules	
	X20 label tabs (blank), 10 unit package	
X20AC0M01.0010	Blank X20 label tabs, white, set for 16 modules, 10 unit package	
X20AC0M02.0010	Blank X20 label tabs, red, set for 16 modules, 10 unit package	
X20AC0M03.0010	Blank X20 label tabs, blue, set for 16 modules, 10 unit package	
X20AC0M04.0010	Blank X20 label tabs, orange, set for 16 modules, 10 unit package	
	X20 label tabs (printed), 10 unit package	
X20AC0M11.0010	Printed X20 label tabs, white, set for 16 modules, 10 unit package	
X20AC0M12.0010	Printed X20 label tabs, red, set for 16 modules, 10 unit package	
X20AC0M13.0010	Printed X20 label tabs, blue, set for 16 modules, 10 unit package	
X20AC0M14.0010	Printed X20 label tabs, orange, set for 16 modules, 10 unit package	

5. Labeling tool

The labeling tool is needed for applying the label tabs.

Model number	Short description	Image
X20AC0MT1	X20 labeling tool for the X20 label tabs	

Chapter 6 • Mechanical design

1. Solid mechanics

With all the advantages that the three-part modularity of the X20 System offers, one emphasis has always been solid mechanical design.

Its robust design, long guides, and strengthened housing guarantee the stability it needs in industrial environments. These features allow the X20 System to be mounted on a rail with the same ease as a rack system. They also make it just as simple to remove it from the rail.

The following sections describe the mechanical design of the X20 System. The following topics are covered:

- Assembly of an X20 System
- Mounting rail installation and removal
- Expanding an X20 System
- Installation of accessories
- Adding the terminal labels
- Terminal block application coding

2. Assembly of an X20 System

There are several ways to assemble an X20 System. Two methods are described below:

Assembly of an X20 System	Description
Method 1	The X20 System is completely assembled and then installed on the mounting rail.
Method 2	The X20 System is installed and assembled directly on the mounting rail.

Table 229: Two of the several methods for assembling an X20 System

Information:

- **Store X20 modules in the protective packaging until immediately before assembly.**
- **Only touch X20 modules on the housing.**
- **Take the necessary protective measures against electrostatic discharges (see also chapter 1 "General information", section 2.3 "Protection against electrostatic discharges" on page 27).**

2.1 Method 1

The X20 System is completely assembled and then installed on the mounting rail.

- 1) Remove X20 modules from protective packaging. Check modules for obvious mechanical damages.
- 2) Insert electronic module in the guides on the bus module.

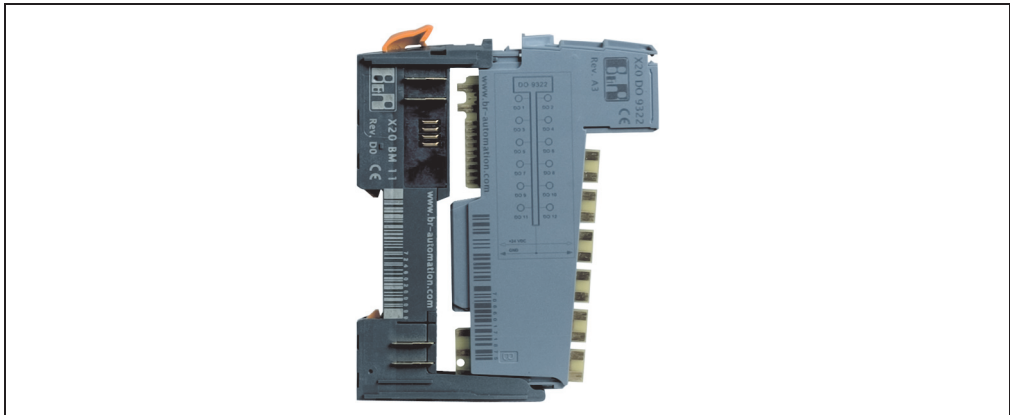


Figure 188: Insert electronic module in the guides on the bus module

- 3) Push the electronic module and the bus module flush together.

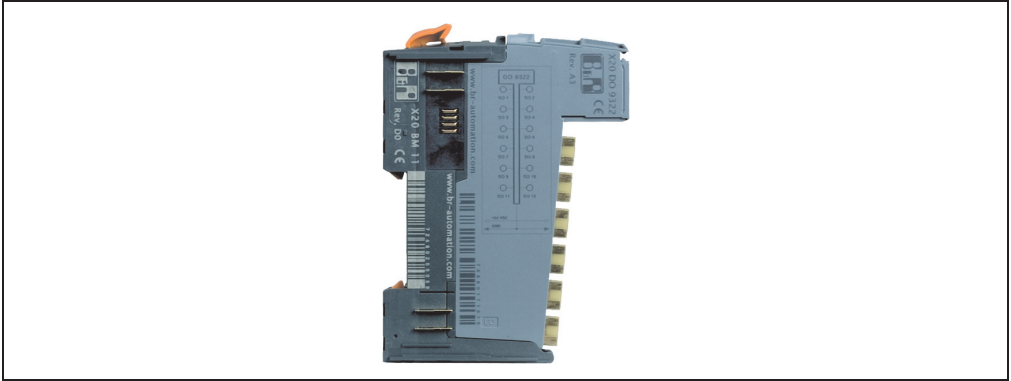


Figure 189: Push the electronic module and the bus module flush together

- 4) Hang the bottom edge of the terminal block in its place on the bus module.

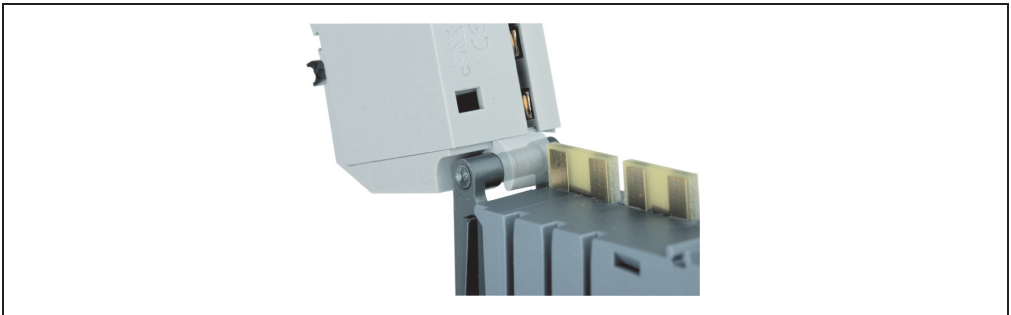


Figure 190: Hang the bottom edge of the terminal block in its place on the bus module

- 5) Rotate the terminal block up into place.

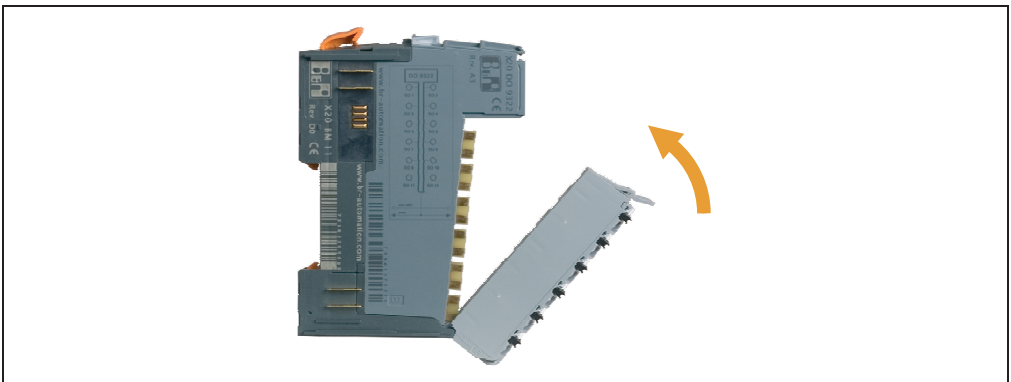


Figure 191: Rotate the terminal block up into place

- 6) The terminal block latch must close with an audible click. If the latch does not catch, the lever must be pushed up.

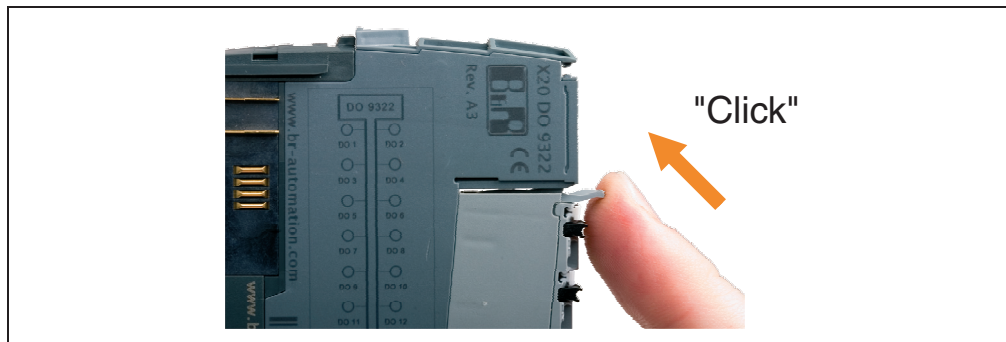


Figure 192: If the latch does not catch, the lever must be pushed up

- 7) Individual X20 modules should be assembled from left to right (viewed from front) to form the complete X20 System. To do this, slide the right module forward into the guides on the left bus module.

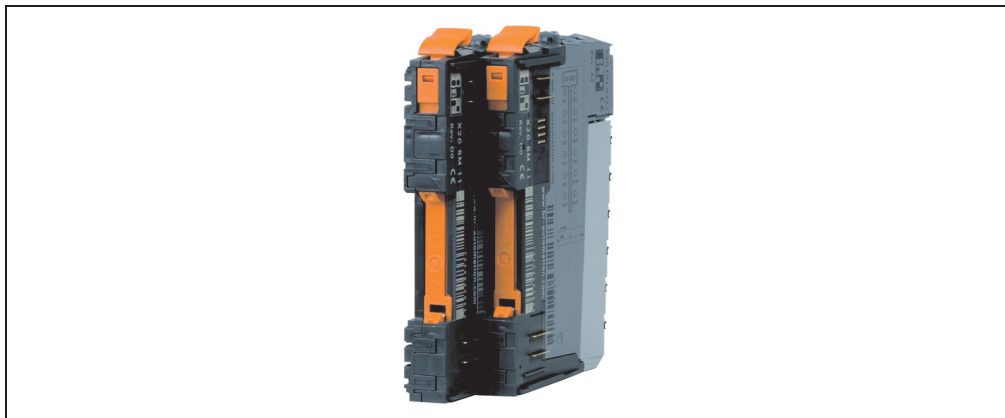


Figure 193: To do this, slide the right module forward into the guides on the left bus module

- 8) Slide the right module forward until the two modules fit flush together.
- 9) Proceed like this until the second to last module.
- 10) For the last module, only insert the bus module in the guides of the left bus module.
- 11) Slide the right bus module forward until the two bus modules fit flush together.

- 12) Insert the right locking plate into the guides on the bus module from the front and push it in all the way in.

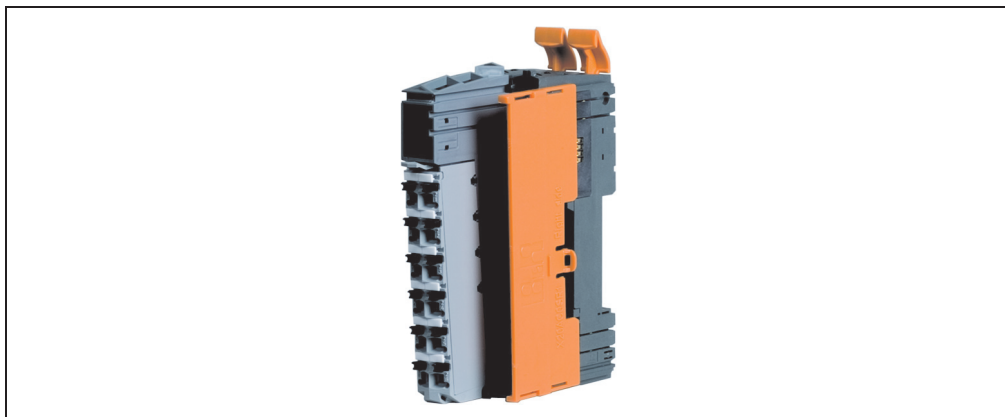


Figure 194: Insert the right locking plate into the guides on the bus module from the front

- 13) Insert the electronic module into the bus module and push firmly so that the two modules fit flush together.
- 14) Hang the bottom of the terminal block in its place on the bus module and push it up into place. The terminal block latch must close with an audible click.
- 15) Lay the left locking plate on the left module and insert it in the guides. Finally, slide the locking plate forward.

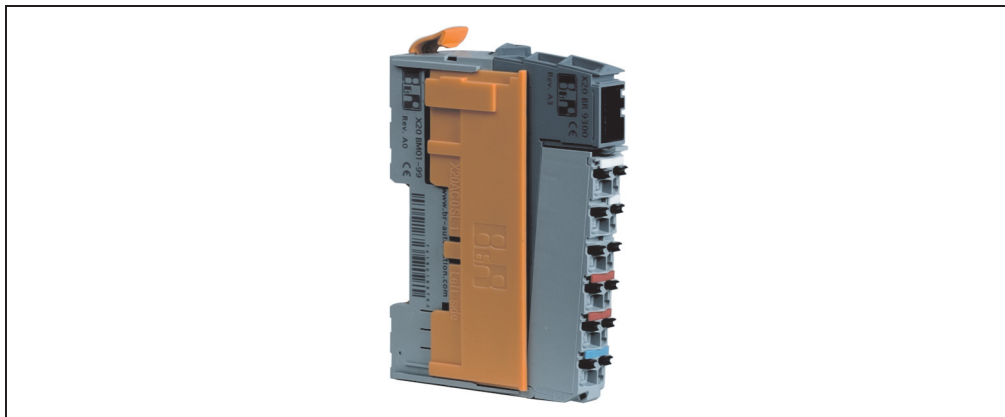


Figure 195: Lay the left locking plate on the left module and insert it in the guides

- 16) The procedure for hanging the X20 system on the mounting rail is described in section 3 "Installing the X20 System on the mounting rail" on page 435.

2.2 Method 2

The X20 System is installed and assembled directly on the mounting rail.

- 1) Remove X20 modules from protective packaging. Check modules for obvious mechanical damages.
- 2) Push the locking lever all the way up on all of the bus modules. This opens the locking mechanism for mounting rail installation.

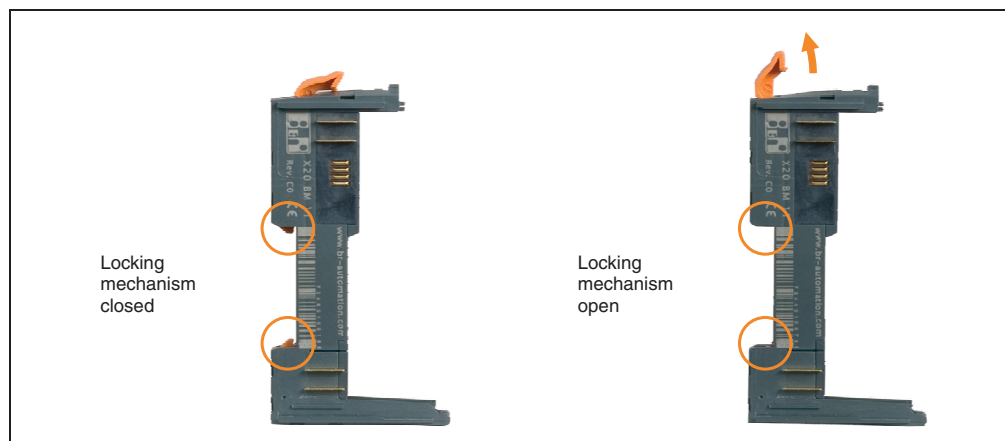


Figure 196: Push the locking lever all the way up to open the locking mechanism

- 3) Hang the first bus module in the desired position on the mounting rail and close the locking mechanism by pushing the lever down.
- 4) Insert the next bus module in the guides of the previously mounted bus module.

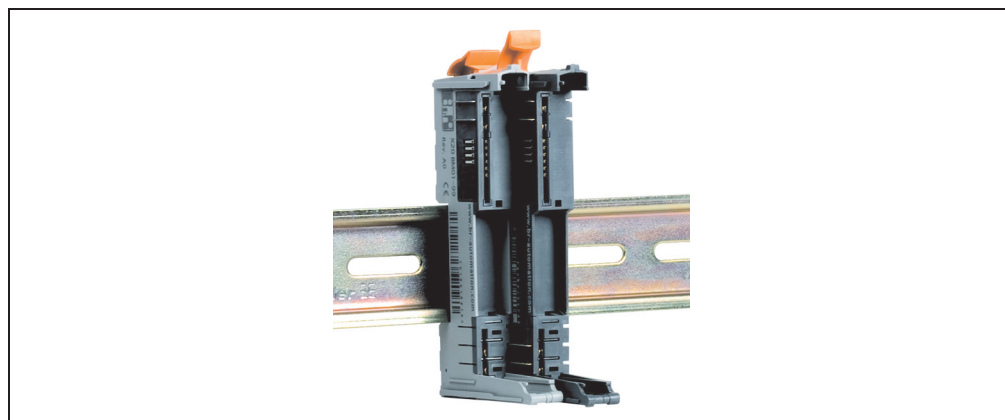


Figure 197: Insert the next bus module in the guides of the previously mounted bus module

- 5) Slide the bus module in against the mounting rail and secure it by pushing down the locking lever.
- 6) Proceed like this with the rest of the bus modules.
- 7) Insert the corresponding electronic module in the guides on the leftmost bus module.

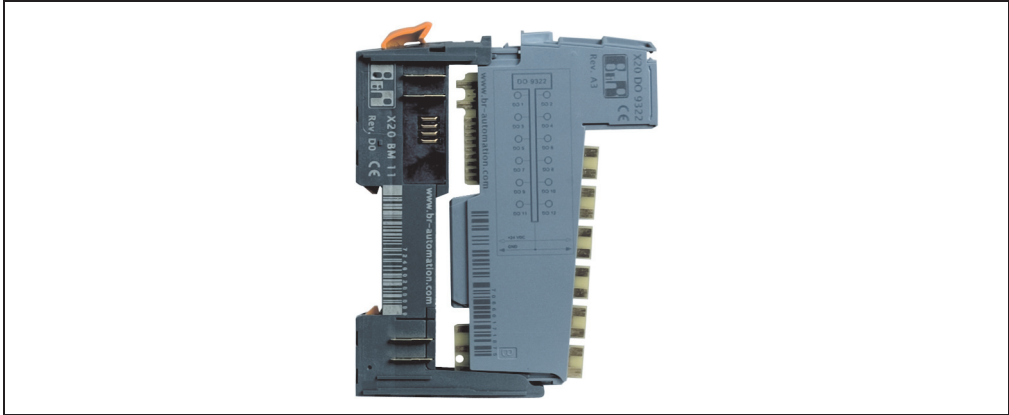


Figure 198: Insert the electronic module in the guides on the leftmost bus module

- 8) Push the electronic module and the bus module flush together.

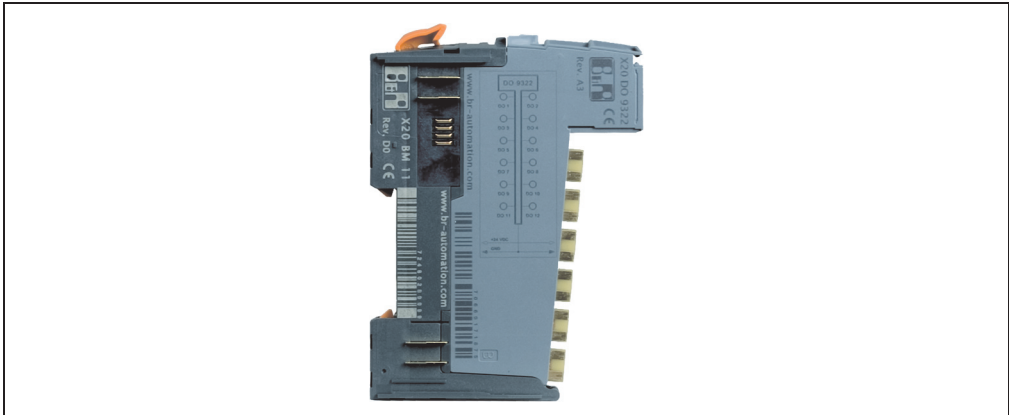


Figure 199: Push the electronic module and the bus module flush together

- 9) Proceed like this until the second to last electronic module.

10) Insert the right locking plate into the guides on the bus module from the front and push it in all the way.

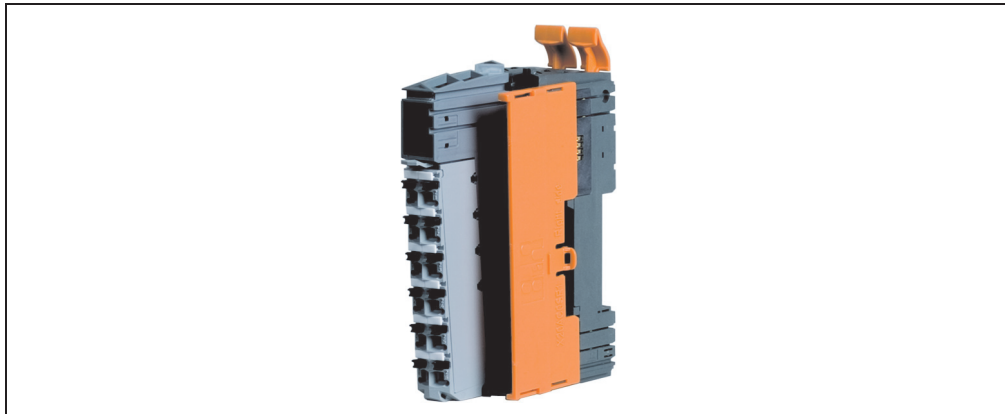


Figure 200: Insert the right locking plate into the guides on the bus module from the front

11) Insert the electronic module into the bus module and push firmly so that the two modules fit flush together.

12) Hang the terminal block in its place on the leftmost bus module.

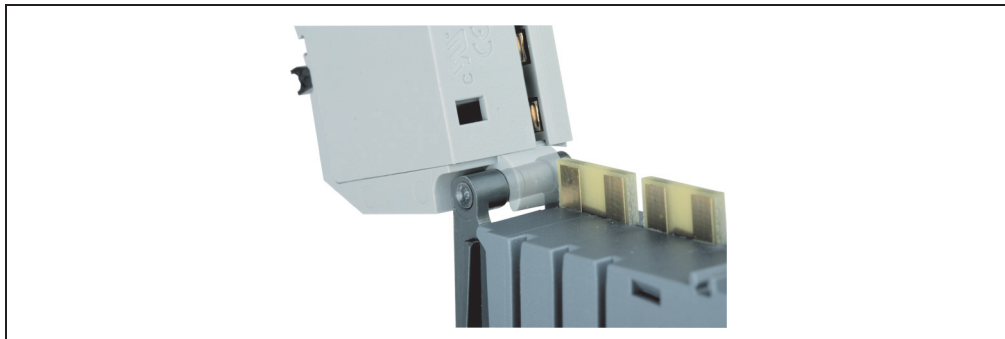


Figure 201: Hang the terminal block in its place on the leftmost bus module

13) Rotate the terminal block up into place.

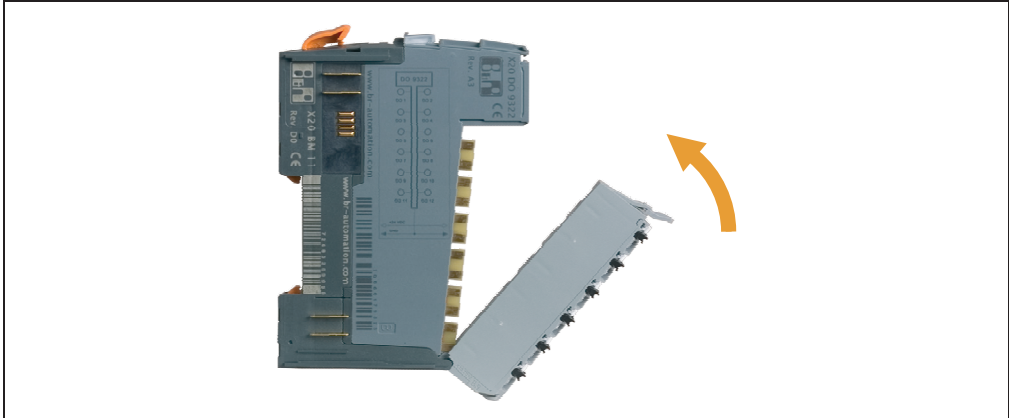


Figure 202: Rotate the terminal block up into place

14) The terminal block latch must close with an audible click. If the latch does not catch, the lever must be pushed up.

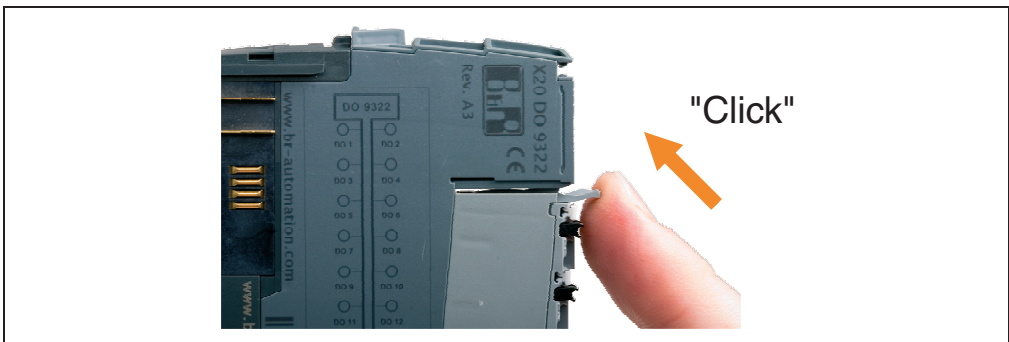


Figure 203: If the latch does not catch, the lever must be pushed up

15) Proceed like this with the rest of the terminal blocks.

- 16) Lay the left locking plate on the left module and insert it in the guides. Finally, slide the locking plate forward.

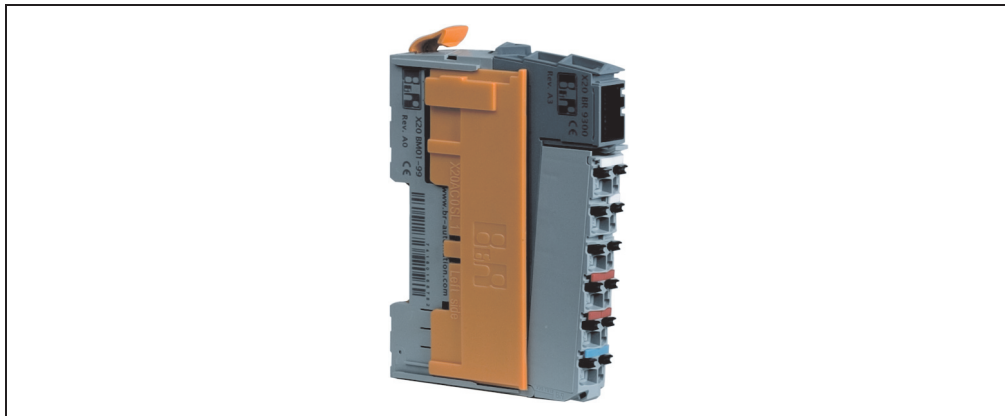


Figure 204: Lay the left locking plate on the left module and insert it in the guides

3. Installing the X20 System on the mounting rail

Complete the following steps to install the X20 System on the mounting rail.

- 1) Push the locking lever all the way up on all of the bus modules. This opens the locking mechanism for mounting rail installation.

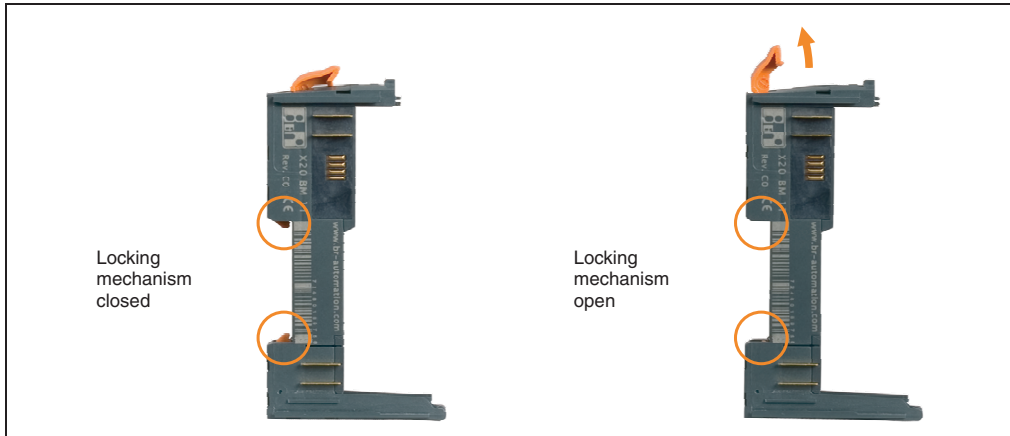


Figure 205: Push the locking lever all the way up to open the locking mechanism

- 2) Hang the X20 System in the desired position on the mounting rail and close the locking mechanism by pushing the lever down.

4. Removing the X20 System from the mounting rail

4.1 Remove the entire system from the mounting rail

- 1) Push the locking lever all the way up on all of the bus modules. This opens the locking mechanism for mounting rail installation.

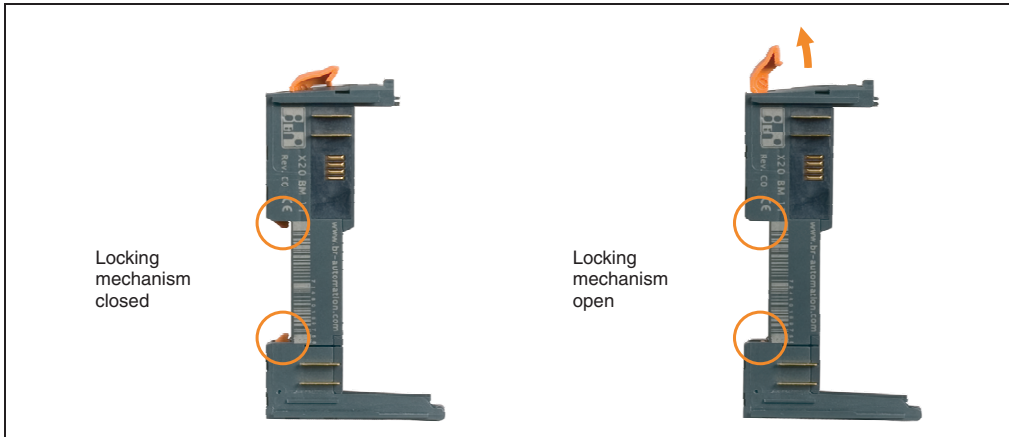


Figure 206: Push the locking lever all the way up to open the locking mechanism

- 2) Remove the X20 System from the mounting rail.

4.2 Remove a block of modules from the mounting rail

- 1) Push the locking lever all the way up on all of the modules that you wish to remove from the mounting rail. This opens the locking mechanism for mounting rail installation.

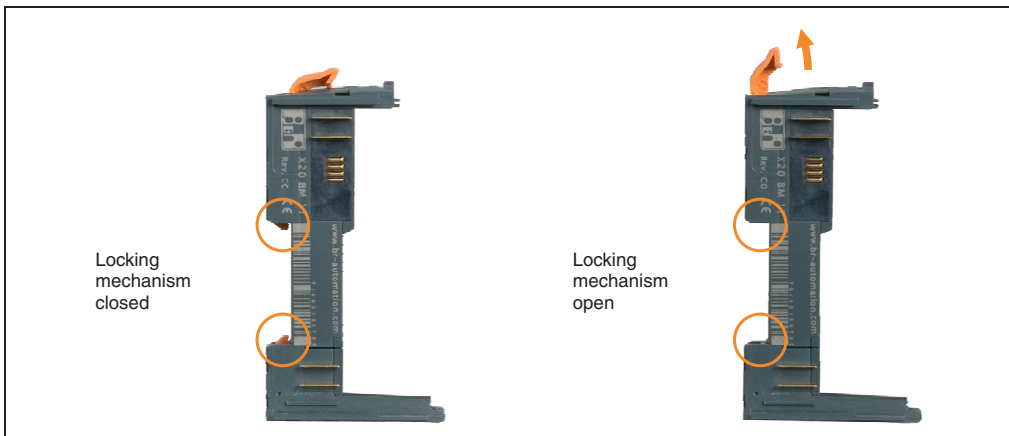


Figure 207: Push the locking lever all the way up to open the locking mechanism

- 2) The terminal block must be removed from the module to the left of the module block that is to be removed. To do this, push down on the locking lever on the terminal block ① and rotate the terminal block out and down ②.

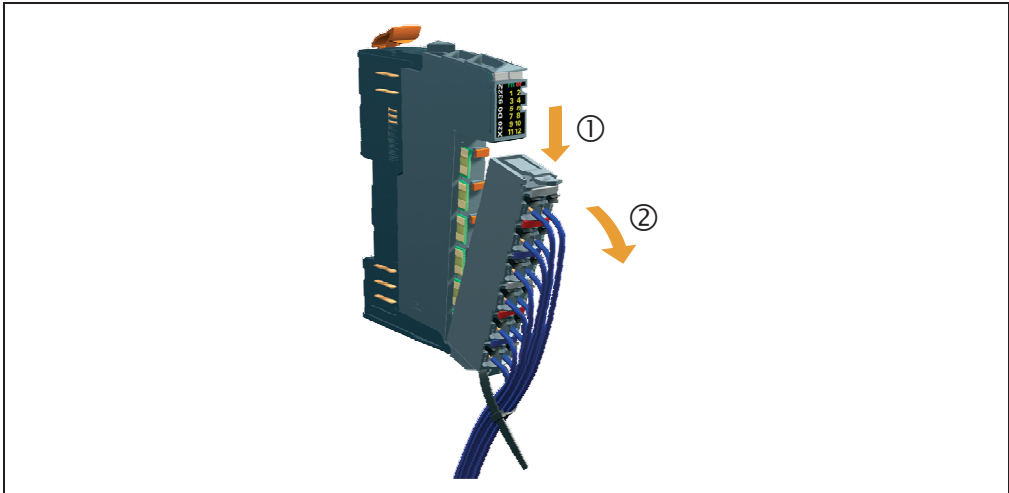


Figure 208: Remove the terminal block from the module to the left

- 3) Remove the module block from the mounting rail.



Figure 209: Module block removed from the mounting rail

- 4) Put the removed terminal block back on the module. To do this, hang the bottom in place in the bus module.

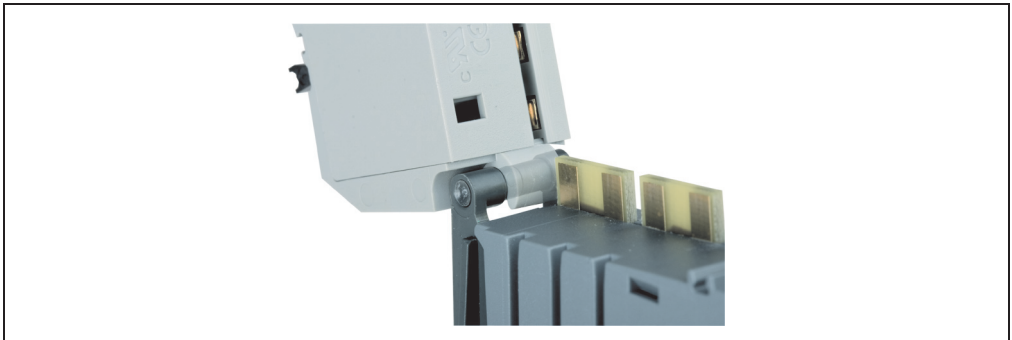


Figure 210: Hang the bottom edge of the terminal block in its place on the bus module

- 5) Rotate the terminal block up into place.

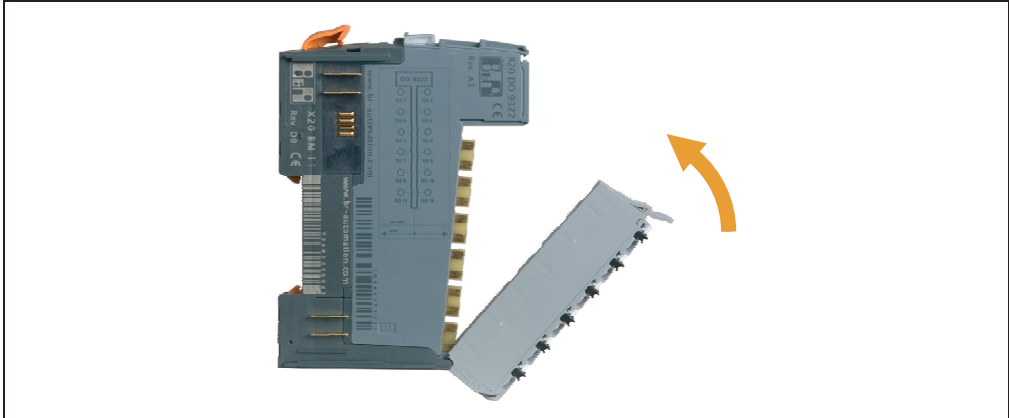


Figure 211: Rotate the terminal block up into place

- 6) The terminal block latch must close with an audible click. If the latch does not catch, the lever must be pushed up.

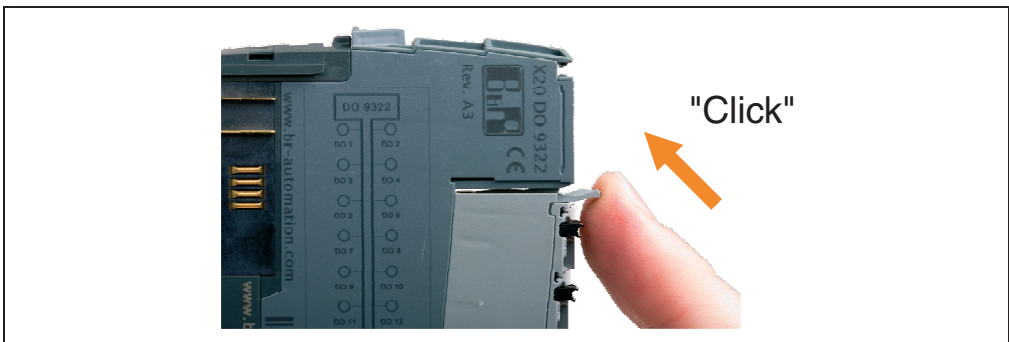


Figure 212: If the latch does not catch, the lever must be pushed up

5. Expanding an X20 System

If you want to expand an existing X20 System to the right, the right locking plate must be removed.

- 1) Remove the terminal block from the rightmost module. To do this, push down on the locking lever on the terminal block ① and rotate the terminal block out and down ②.

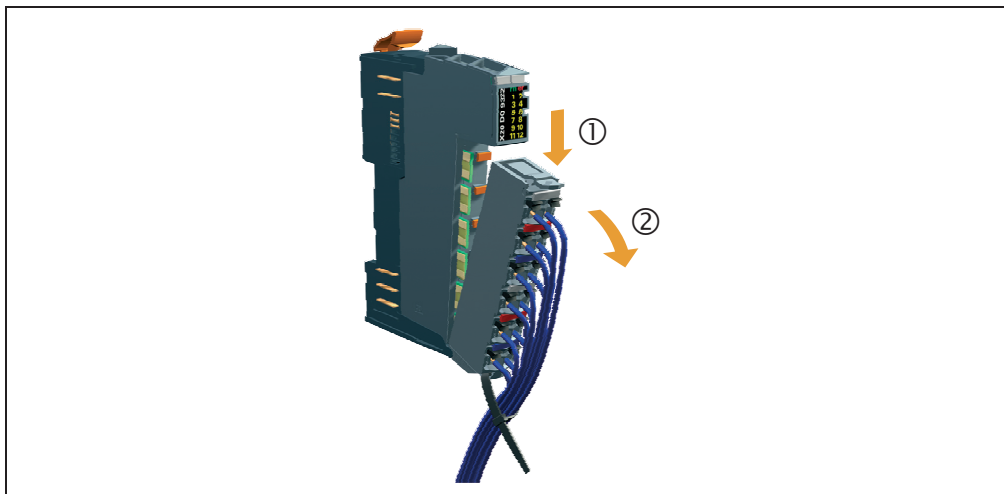


Figure 213: Unplug the terminal block on the rightmost module

- 2) Push down on the electronic module's locking lever ① and remove the electronic module ②.

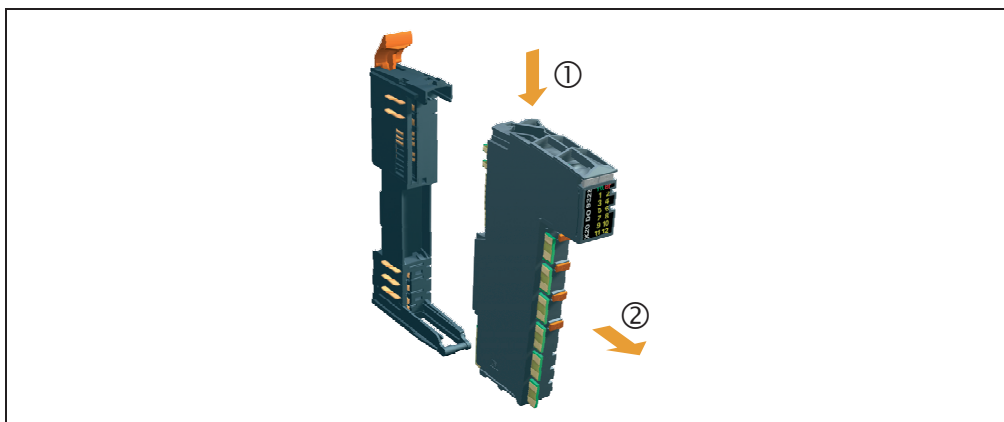


Figure 214: Remove the electronic module